

Comparative Analysis of Methods for Performing a Side-Channel Video-Fingerprinting Attack

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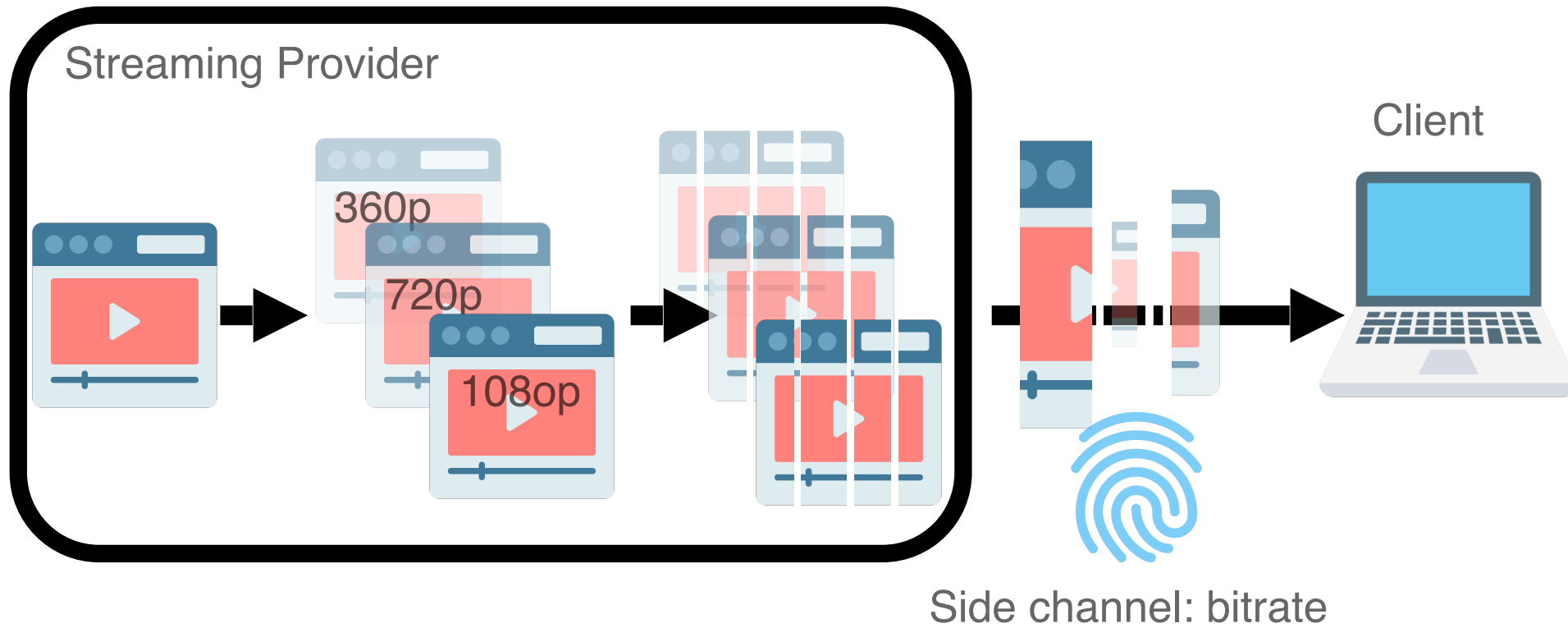


Outline



Background

Dynamic Streaming over HTTP (DASH)



Background

Threat Model

Attacker

Fingerprint
Database

CNN, DTW

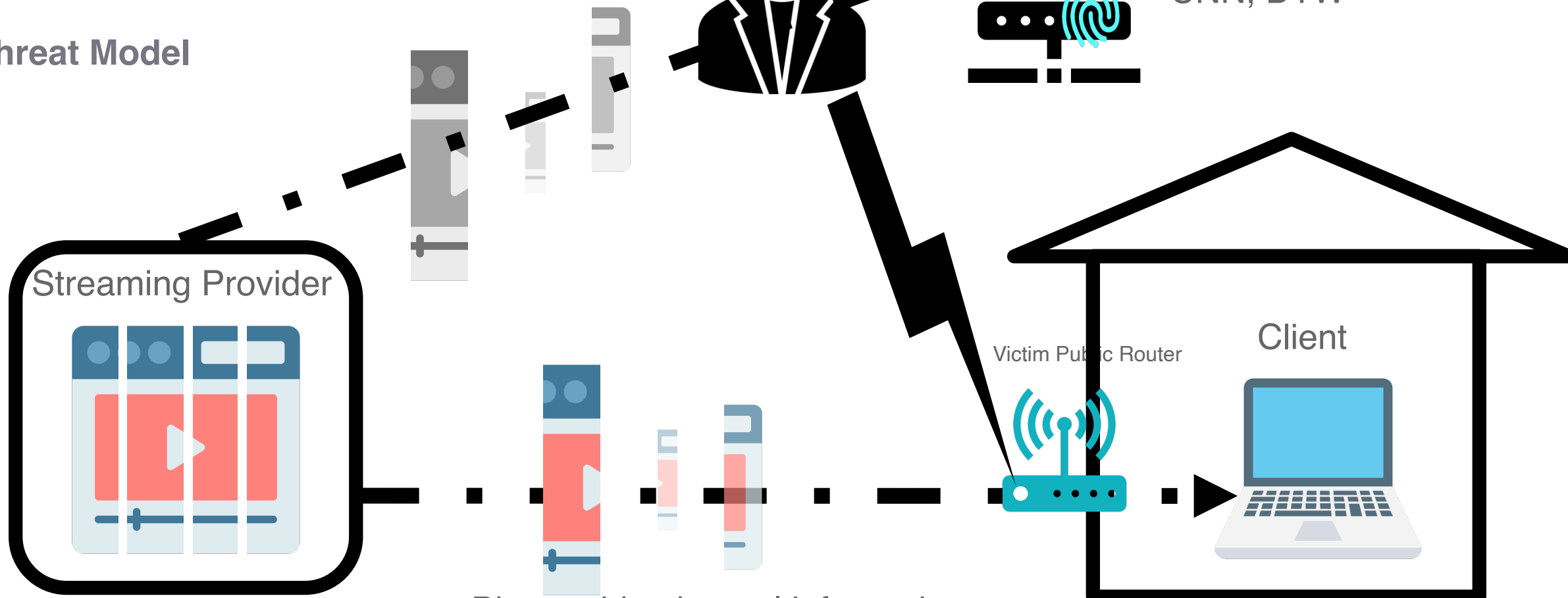
Streaming Provider

Victim Public Router

Client

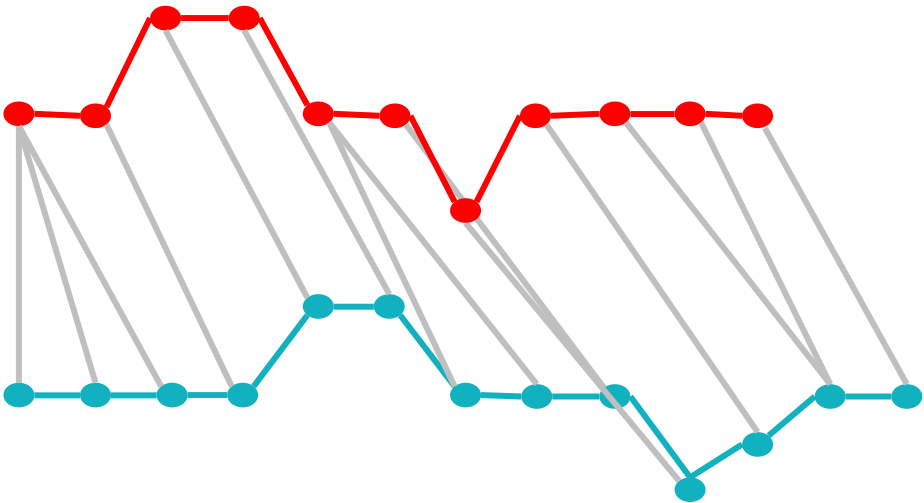
Bitrate side-channel information

[10], [11]



Background

- Dynamic Time Warping (DTW)
 - Distance measure for comparing two time series
 - Edit distance



Background

Threat Model

Attacker

Fingerprint
Database

Closed World

CNN, DTW, **LG**

Dataset Shifts

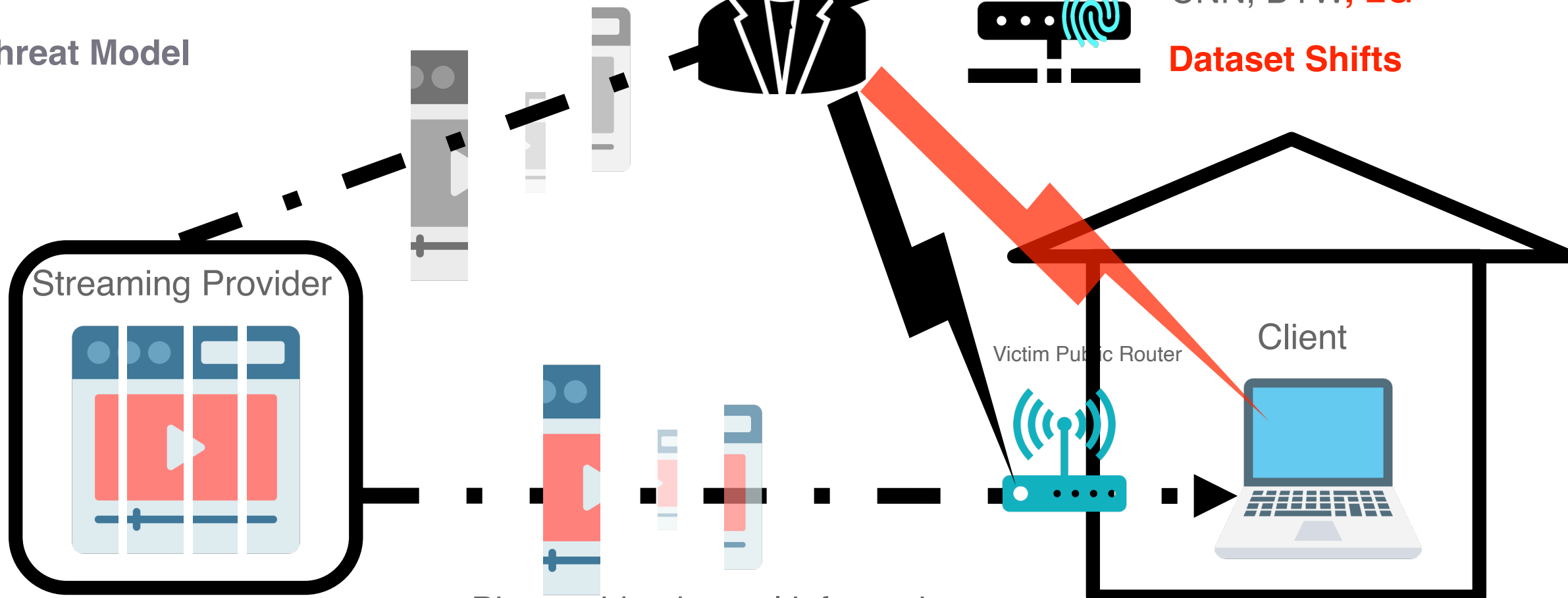
Streaming Provider

Client

Victim Public Router

Bitrate side-channel information

[1], [12], [13]



Research Questions

- To what extent is there an identifiable pattern in HTTPS packet data from YouTube streams?
- How well does a CNN perform at predicting YouTube videos in an open-world scenario using bitrate information compared to the less complex methods of logistic regression and a DTW-based model?

Dataset

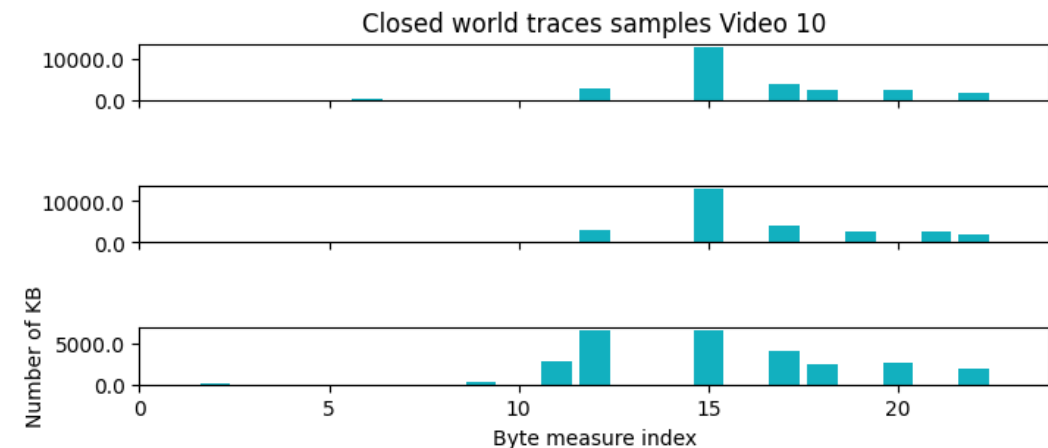
- 6289 HTTPs-packet size records
- 2 minutes of random YouTube videos
- Automated data capture with Selenium
- $50 \times 50 = 2500$ Closed-World traces
- Remaining Traces are Open-World traces

Dataset sample

index	0
timestamps	[1563289706.330096, 1563289706.330207, 1563289706.330488, 1563289715.716338, 1563289716.346138,...]
sizes	[120, 88, 120, 88, 120, 88, 120, 120, ...]
Video ID	Video 10

Predictors → timestamps, sizes

Outcome → Video ID

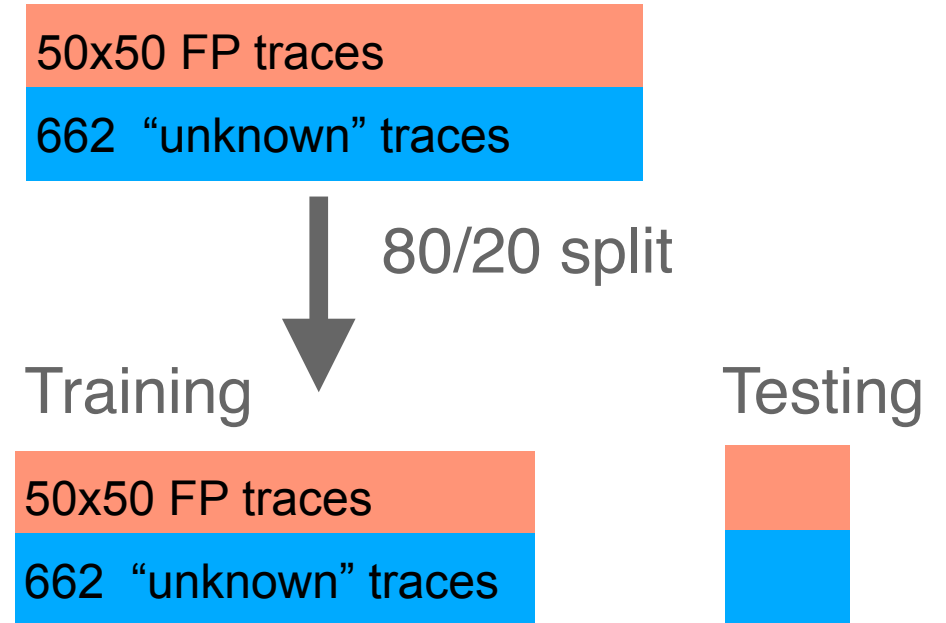


[1], [12], [13]

Dataset

- 6289 HTTPs-packet size records
- 2 minutes of random YouTube videos
- Automated data capture with Selenium
- $50 \times 50 = 2500$ fingerprinting traces
- Remaining traces are labelled “unknown”

Closed-World Dataset



Dataset

- 6289 HTTPs-packet size records
- 2 minutes of random YouTube videos
- Automated data capture with Selenium
- 50 x 50 = 2500 fingerprinting traces
- Remaining traces are labelled “unknown”
- Evaluation metric:

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

Open-World Dataset

50x50 FP traces

662 “unknown” traces

Training

50x50 FP traces

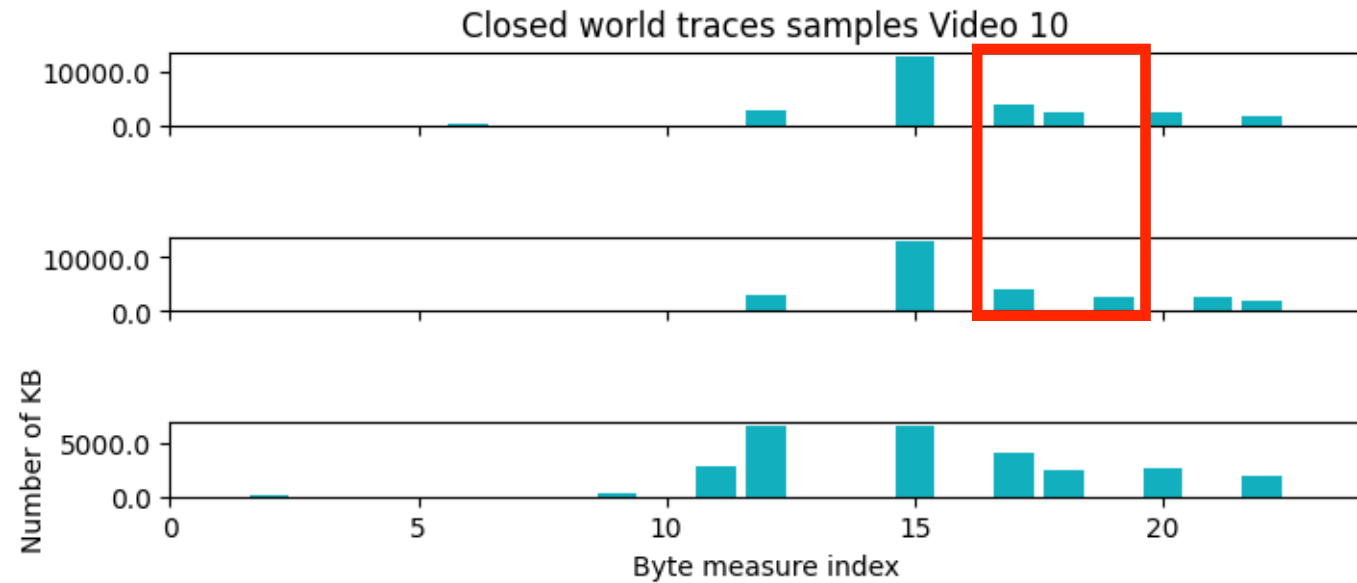
662 “unknown” traces

Testing

3000
“unknown”
traces

Dataset

- Dataprocessing
 - Aggregation into bytes per second
 - Cropping into 2-minute intervals

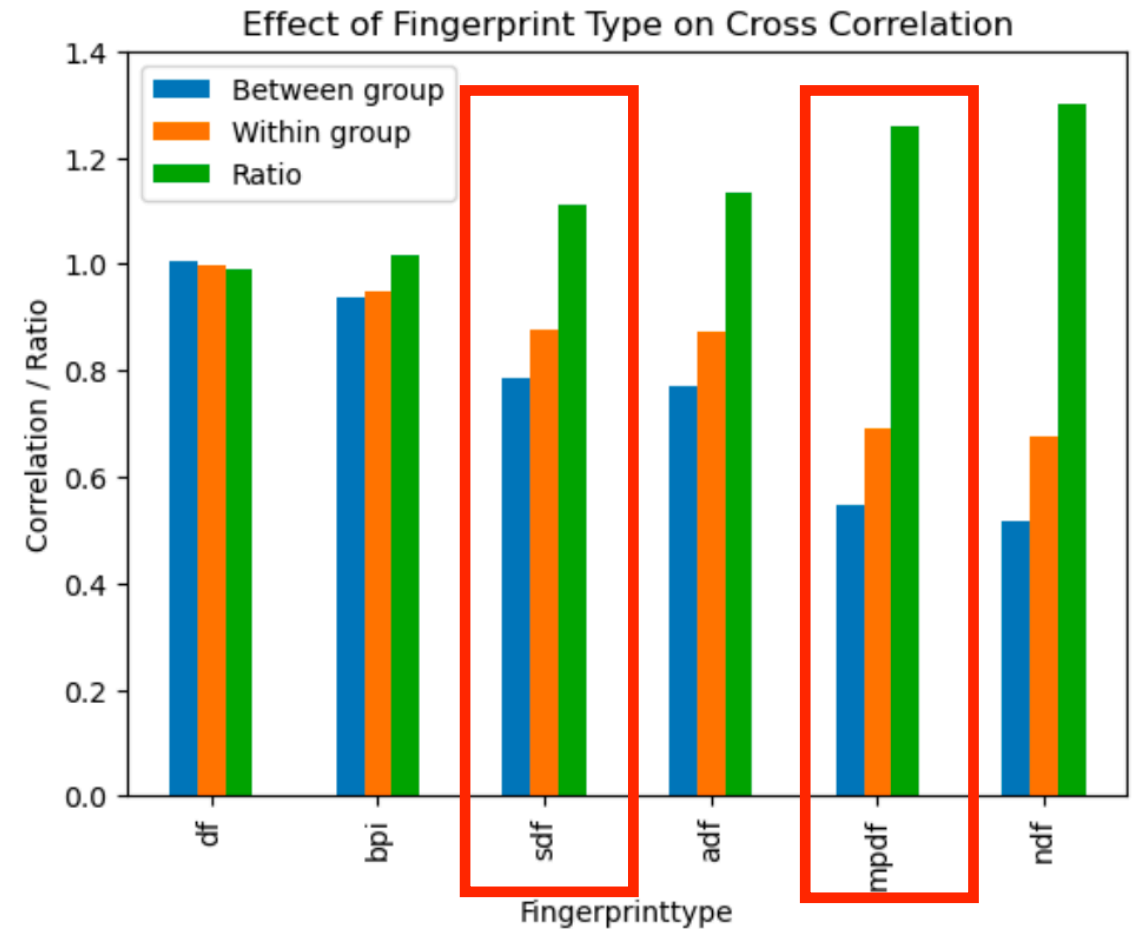


Dataset

- Dataprocessing
 - Aggregation into bytes per second
 - Cropping into 2-minute intervals
- Fingerprinting formula

Cross Correlation

$$R_{s_i s_j}(\tau) = \sum_t s_i(t) \cdot s_j(t + \tau)$$



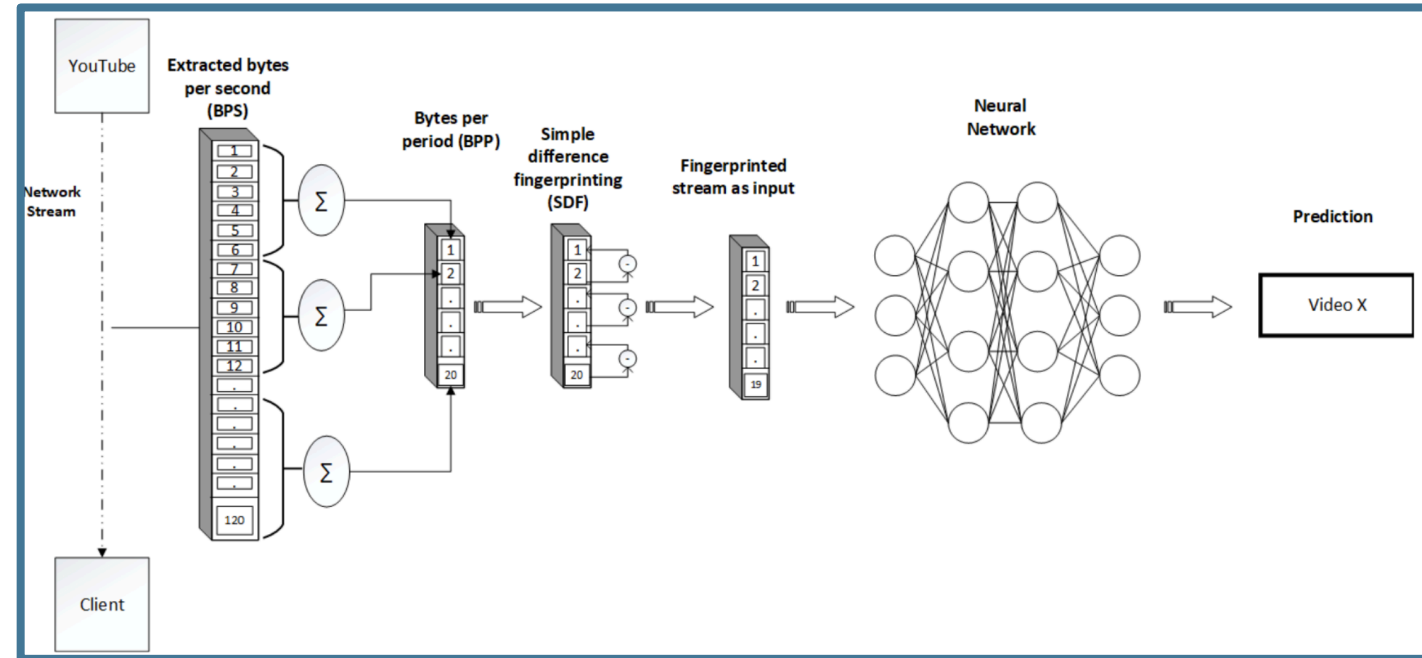
$$f_i = F_{sdf}(s_i, s_{i-1}) = s_i - s_{i-1}$$

$$f_i = F_{mpdf}(s_i, s_{i-1}) = \frac{s_i - s_{i-1}}{\max(s_i, s_{i-1})}$$

Modelling

Convolutional Neural Network

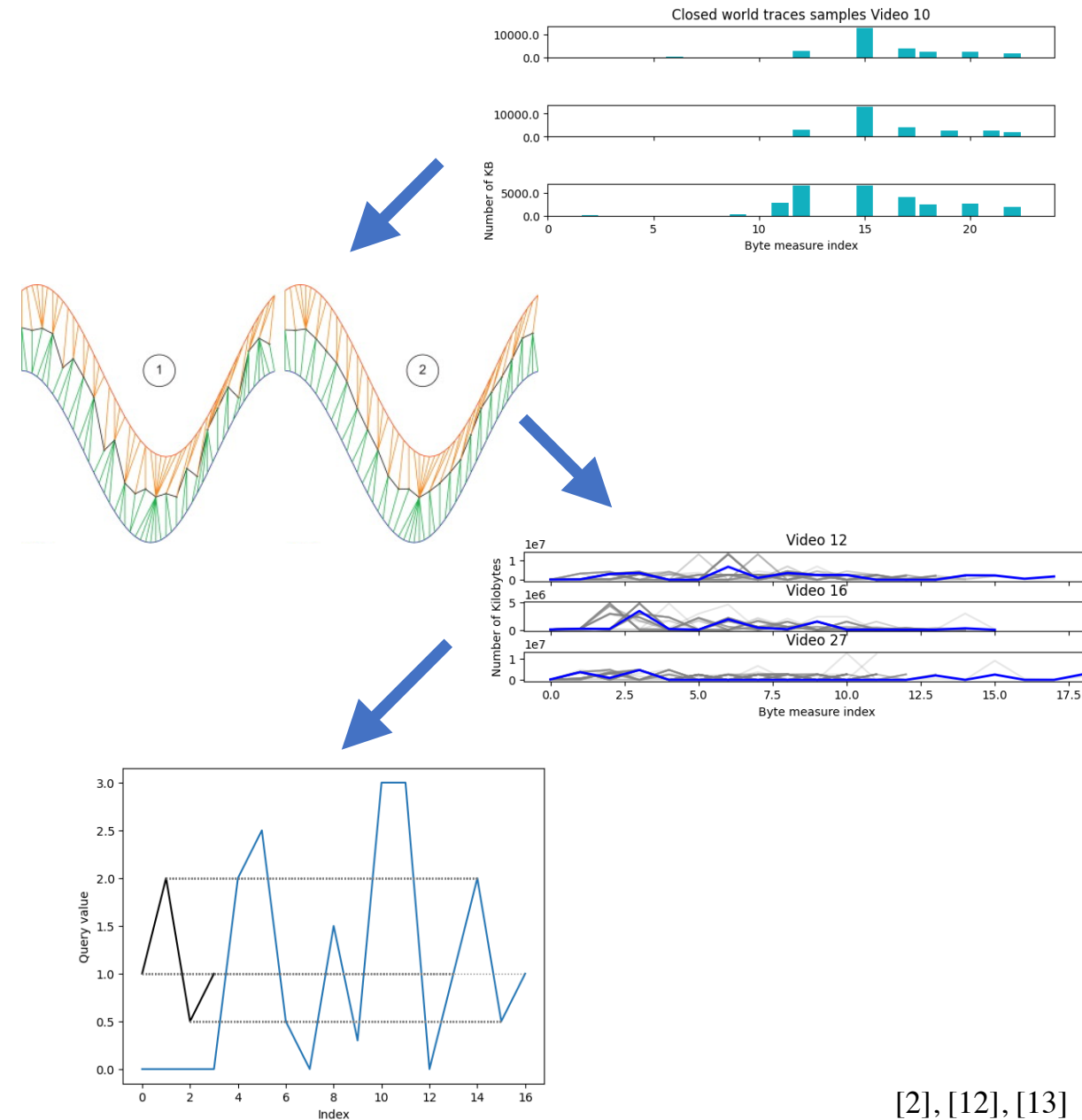
- Follow [3]
- SDF and 0 padding
- Increased drop-out rate $0.2 \Rightarrow 0.9$
- Reduced Epochs $500 \Rightarrow 100$



Modelling

Dynamic Time Warping

- Adopt [2] Partial DTW model
- Feature Engineering
 - Calculate barycentre average of 10 best traces
 - Length
 - Variance
 - averages of most most ideate



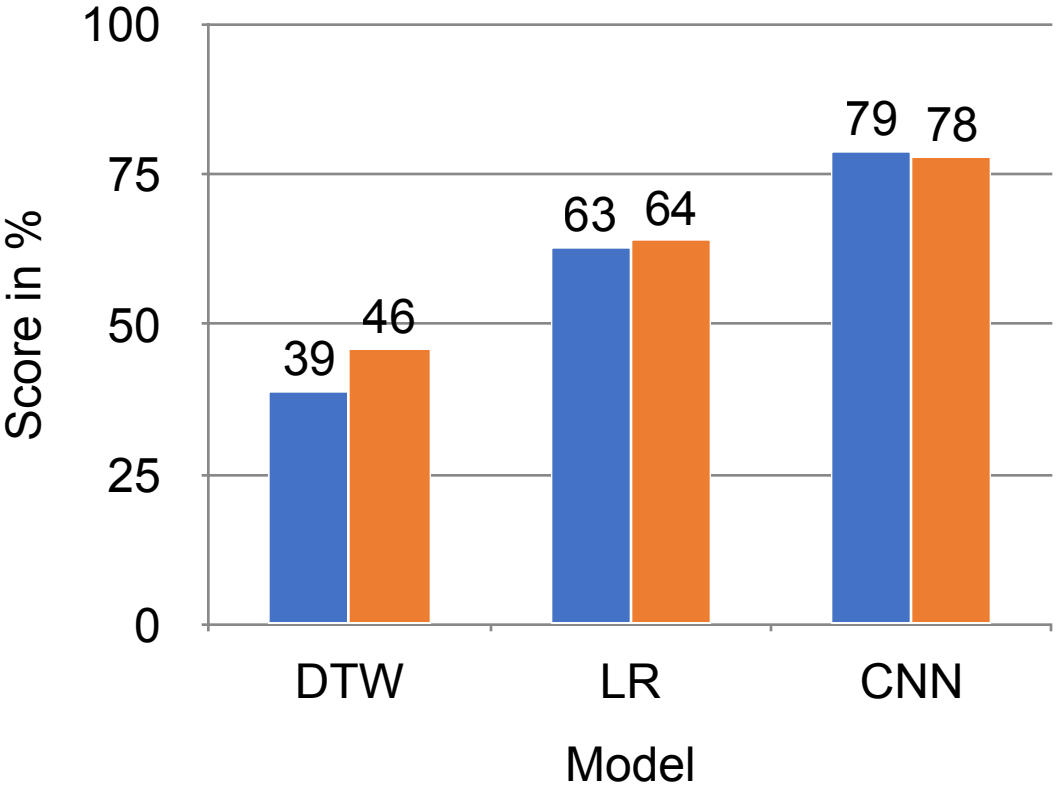
Modelling

Logistic Regression

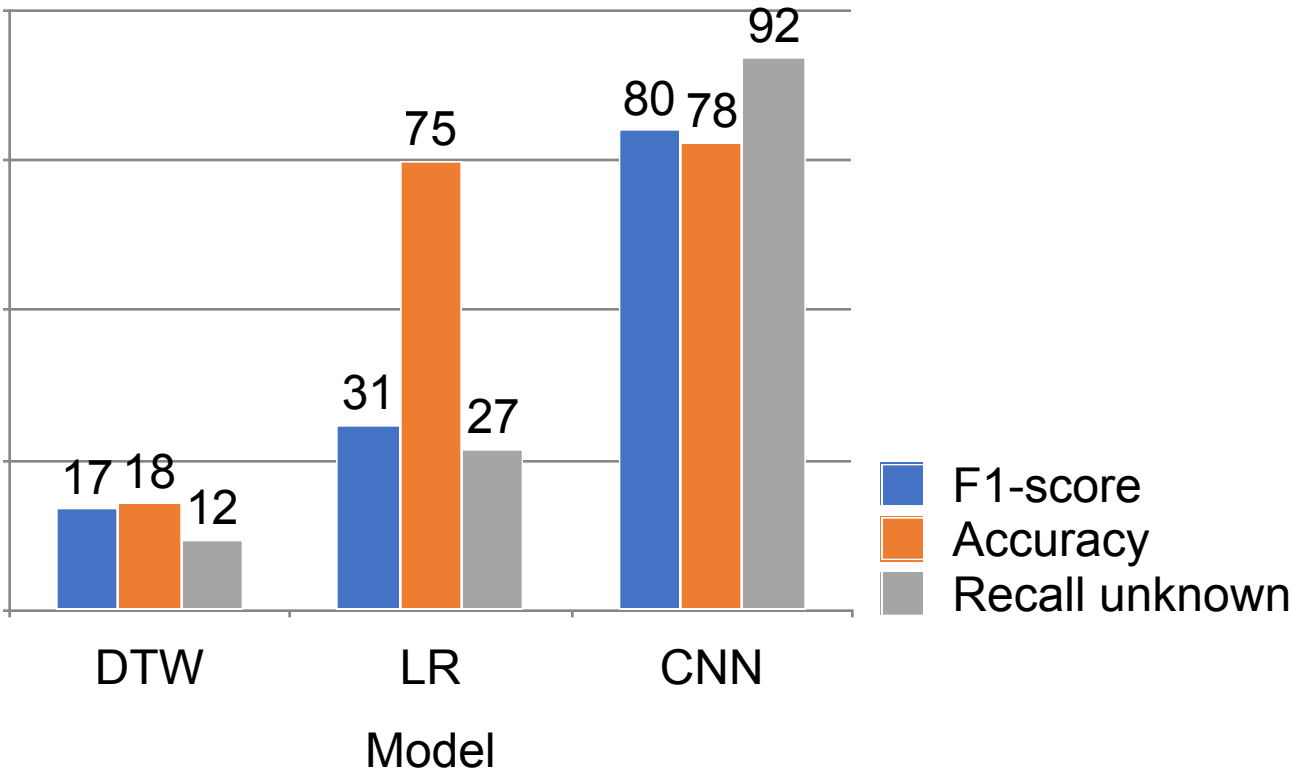
- TSfresh: automatic time-series feature extraction and selection
- One-vs-rest logistic regression strategies [22]
- 0.2 “unknown” threshold
- Multicollinerety, feature reduction, correlation
- L2 Regularisation

Results

Closed World



Open World



Conclusions and Future Works

- SCA for encrypted video-traffic-fingerprinting using less granular data than previous authors is possible
- Found extractable and identifiable patterns in high-level network traces with all proposed models.
- CNN model performed best in open and closed-world scenarios
- Less complex models Logistic Regression and DTW failed to generalise on unrelated samples and are not suitable for real-world implementation.

Conclusions and Future Works

- Compare and examine more algorithms for video-traffic fingerprinting.
- Explore additional methods to protect against SCNN-attacks on encrypted network traces.
- Utilise the extensive dataset of over 6000 open and closed-world traces made available for further research.

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Questions?

Presentation	Deborah Djon	Darragh Connaughton
		